Agent 3: Quadratic Solution

From $w^2 - 12w + 32 = 0$
Using quadratic formula or factoring: $(w - 8)(w - 4) = 0$
So $w = 8$ or $w = 4$
Dimensions: 8×4
Answer: 8 units by 4 units

Final round:

Agent 1: Adopts Simpler Method

"Agent 2's approach of finding factors of 32 that sum to 12 is much more direct than solving the quadratic. Since $8 \times 4 = 32$ and $8 + 4 = 12$, the dimensions are 8 by 4."
Recognizes and adopts more efficient solution method
Answer: 8 units by 4 units

Key observation. Agent 1 explicitly acknowledges that Agent 2's factor-finding approach is "much more direct" than the algebraic method, demonstrating recognition of solution efficiency. Post-training reinforces these behaviors, enabling the model to learn patterns of concise and efficient reasoning.

B.4 Summary

These examples demonstrate observable grounding behaviors in multi-agent debates:

Error propagation: Agents sometimes abandon partially correct reasoning to adopt peer errors (Example 1, Agent 1).

Explicit error recognition: Agents identify specific errors in their own reasoning ("I added the discount instead of subtracting") and in peers' approaches ("adding time to flight departure").

Method adoption: Agents recognize more efficient solution methods from peers and explicitly acknowledge this ("Agent 2's approach... is much more direct").

Self-correction through peer comparison: Seeing consistent peer answers prompts agents to revisit and correct their calculations.

These behaviors generate training signals beyond simple majority voting. The debate format captures not just correct answers but also common error patterns, self-correction mechanisms, and explicit reasoning about solution quality, providing rich supervision for teaching models to recognize and adopt sound reasoning strategies.

C MACA Algorithm: Additional Implementation Details

C.1 Agent Heterogeneity and Specialization

In our current experiments, we use homogeneous agents (clones) that update independently over a single training iteration, enabling evaluation of both individual and averaged performance to measure the benefits of divergence. We also explore the impact of training over multiple such iterations of debate and post-training in App. G. For multi-iteration runs, agent checkpoints can be managed in two ways: agents may either remain synchronized by resetting checkpoints to the best performing one after every iteration, or diverge by updating independently from distinct trajectory data, allowing specialization and diversity through differential learning. Additionally, the MACA framework also supports heterogeneous agents, allowing different language models to be independently optimized through this framework, where agents represent distinct models or architectures from the outset, each bringing inherently different capabilities or perspectives.

Algorithm 1 Multi-Agent Consensus Alignment Via Debate-RL

input: agents $\{\pi_{\theta_m}\}_{m=1}^{M}$, rounds R, iters L, batch size B, method in MV-SFT, MV-GRPO, MV-DPO, or MV-KTO
for $\ell = 1, \ldots, L$ **do**
  sample batch $\{x^{(\ell,i)}\}_{i=1}^{B}$ of prompts from $\mathcal{D}$
  for $i = 1, \ldots, B$ **do** ▷ Generate debate trajectories
    $y_{m,1}^{(\ell,i)} \sim \pi_{\theta_m}(\cdot \mid x^{(\ell,i)}) \quad \forall m$
    for $r = 2, \ldots, R$ **do** ▷ Condition on peers' previous round outputs
      $x_{m,r}^{(\ell,i)} = [x^{(\ell,i)}; \{y_{j,r-1}^{(\ell,i)}\}_{j \neq m}]$
      $y_{m,r}^{(\ell,i)} \sim \pi_{\theta_m}(\cdot \mid x_{m,r}^{(\ell,i)}) \quad \forall m$
    end for
    Parse final answers $a_m^{(\ell,i)} = \mathcal{A}(y_{m,R}^{(\ell,i)}) \quad \forall m$
    Aggregate consensus $\hat{a}(x^{(\ell,i)}) = \text{Majority}\{a_1^{(\ell,i)}, \ldots, a_M^{(\ell,i)}\}$
    Extract final contexts $x_m^{(\ell,i)} \leftarrow x_{m,R}^{(\ell,i)}$ for all m
    Partition responses by consensus alignment:
      $\mathcal{G}_+^{(\ell,i)} = \{y_{m,R}^{(\ell,i)} : a_m^{(\ell,i)} = \hat{a}(x^{(\ell,i)})\}$
      $\mathcal{G}_-^{(\ell,i)} = \{y_{m,R}^{(\ell,i)} : a_m^{(\ell,i)} \neq \hat{a}(x^{(\ell,i)})\}$
  end for
  build dataset: $\mathcal{D}_{\text{post}} = \{(x^{(\ell,i)}, \hat{a}(x^{(\ell,i)}), \mathcal{G}_+^{(\ell,i)}, \mathcal{G}_-^{(\ell,i)})\}_{i=1}^{B}$
  training data by method:
    MV-SFT: for each m, use $\{(x_m^{(\ell,i)}, y_{m,R}^{(\ell,i)}) : y_{m,R}^{(\ell,i)} \in \mathcal{G}_+^{(\ell,i)}\}_i$.
    MV-GRPO: for each m, store $\{(x_m^{(\ell,i)}, \hat{a}(x^{(\ell,i)}))\}_i$ for reward computation on new samples.
    MV-DPO/MV-KTO: pool $\{(x_m^{(\ell,i)}, y_{m,R}^{(\ell,i)})\}_{i,m}$ partitioned by $\mathcal{G}^+, \mathcal{G}^-$ across agents.
  update policies: apply majority-vote objective to refine each π_{θ_m}
end for

D Experimental Setup

D.1 Training Parameters

We conducted hyperparameter sweeps across learning rates from 1e-7 to 1e-5, finding that 1e-5 consistently performed best across all methods. For preference-based methods (MV-KTO, MV-DPO), we used $\beta = 0.1$ throughout all experiments. LoRA ranks and alpha values were adjusted based on model size and computational constraints, with rank/alpha of 64-128 for MV-DPO and MV-KTO, 32-64 for MV-GRPO, and 128 for MV-SFT. Tables 7–10 provide complete parameter specifications.

GSM8K, MATH, and MathQA used 1500/500/500 train/valid/test splits. SVAMP, GPQA, CSQA, and AMC were used only for testing, with 300, 448, 500, and 40 test examples, respectively.

D.2 Datasets

We evaluated model mathematical reasoning using seven publicly available datasets spanning a range of difficulty and subject areas: MATH (Hendrycks et al., 2021), GSM8K (Cobbe et al., 2021), MathQA (Amini et al., 2019), SVAMP (Patel et al., 2021), GPQA (Rein et al., 2023), AMC 23 kno (2025), and CommonsenseQA/CSQA (Talmor et al., 2019).

MATH (Hendrycks et al., 2021): The MATH dataset contains 12,500 high school mathematics problems from algebra, geometry, combinatorics, and number theory, each requiring multi-step reasoning and precise mathematical solutions.

GSM8K (Cobbe et al., 2021): GSM8K is composed of 8,500 grade-school-level word problems emphasizing arithmetic and logical reasoning, with step-by-step annotated solutions for each problem.

MathQA (Amini et al., 2019): MathQA features over 37,000 question-answer pairs based on quantitative reasoning, requiring models to convert natural language problems into mathematical expressions and perform multi-step computation.

SVAMP (Patel et al., 2021): SVAMP consists of carefully reworded arithmetic word problems designed to probe model robustness and prevent reliance on annotation artifacts, focusing on genuine multi-step arithmetic reasoning.

Table 7 Training parameters.

Parameter	SFT	GRPO	DPO	KTO
Learning rate	$1 \times 10^{-5\text{–}7}$	$1 \times 10^{-5\text{–}7}$	$1 \times 10^{-5\text{–}7}$	$1 \times 10^{-5\text{–}7}$
Weight decay	1×10^{-2}	1×10^{-2}	1×10^{-2}	1×10^{-2}
Batch size	1–8	8	1–8	1–8
Epochs	1–3	1–3	1–3	1–3
Gradient accumulation steps	2–4	2–4	2–4	2–4
Optimizer	AdamW	AdamW	AdamW	AdamW
LoRA rank (r)	8–128	8–64	8–128	8–128
LoRA alpha	8–128	8–64	8–128	8–128
LoRA dropout	0.1	0.1	0.1	0.1
LoRA target modules	q_proj, k_proj, v_proj, o_proj			
Entropy coefficient	-	0.01	-	-
Beta (regularization)	-	-	0.1–0.3	0.1–0.3
Number of generations per step	-	8	-	-
Maximum sequence length	Model-dependent (2048–4096)			
Mixed precision	bf16			
Quantization	4-bit (BitsAndBytesConfig)			

Table 8 Generation parameters.

Parameter	Value
Temperature	1.0
Top-p sampling	0.9
Maximum new tokens	256
Do sample	True
Pad token ID	EOS token ID

GPQA (Rein et al., 2023): GPQA provides 448 expert-curated, multiple-choice questions covering graduate-level biology, physics, and chemistry, emphasizing deep, multi-stage reasoning and robust factual understanding.

AMC 23 (kno, 2025): This dataset includes recent problems drawn from the 2023 American Mathematics Competitions (AMC), spanning algebra, combinatorics, geometry, and number theory, and is useful for assessing model performance on expert-constructed math tasks

CommonsenseQA (Talmor et al., 2019): CommonsenseQA is a challenging multiple-choice question-answering dataset requiring models to apply commonsense reasoning over diverse everyday scenarios. The dataset comprises 12,247 questions, each designed to probe deeper, non-trivial conceptual knowledge beyond factual recall, making it a benchmark for evaluating commonsense understanding in language models.

These datasets collectively enable a thorough evaluation of analytical reasoning capabilities in large language models.

D.3 Multi-Agent Debate Infrastructure

Our multi-agent debate system addresses the computational challenges of training and deploying multiple agents efficiently through specialized infrastructure for quantized training isolation and scalable debate inference.

D.3.1 Quantized MARL Training

Current post-training libraries lack support for multi-gpu distributed training with 4-bit quantization and PEFT adapters. We implement training isolation where each agent trains on a single GPU with explicit resource assignment, using